Math for Data Science - Lab 7

Course: Math for Data Science

This lab is about **Bayesian inference** with the beta-binomial model: combining a prior belief about a probability θ with observed data to obtain a posterior distribution. We derive posteriors by hand, meet **conjugacy**, derive the mean and variance of the beta distribution, and then apply the machinery to a real dataset on placenta previa births. A final optional exercise proves two general results on conditional mean and variance.

Exercises 1-3 are the core of the lab; within them only the code questions at the end of exercise 1 (questions 3 and 4) are *[optional]*. Exercise 4 is *[optional]* as a whole.

Every question is marked with what kind of answer it wants:

- **[Code]** - write the code in the cell marked `# here your code`, then run it.
- **[Pen & paper]** - a calculation, derivation or explanation. Write it in the cell marked *here your solution* (or on paper).
- **[Code + Pen & paper]** - write the code, then explain what you see in the text cell below.

Setup

Run the cell below once to import everything we need.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy import stats
```

Using R instead of Python (optional)

The lab is written in Python, but you can write your answers in R inside this same notebook. This needs R 4.5 or newer, and Jupyter started with rpy2 added: `uv run --with rpy2 jupyter lab`. The `README.md` of the course folder has the details and what to do when it does not work.

If you want to use R instead of Python, uncomment and run the following cell:

```
# %load_ext rpy2.ipython
```

Then, in any subsequent code cell, add `%%R` as the first line to run that cell as R:

```
%%R
x <- rbinom(1000, size = 10, prob = 0.3)
hist(x)
```

Keep running the given Python cells (imports, data loading) as they are. The hints name Python functions, such as `rng.binomial`; you will need to find the R equivalent (`rbinom`) yourself.

Exercise 1: Posterior Distribution

Questions 1 and 2 are done by hand. Questions 3 and 4 check the results in code and are *[optional]*.

As a reminder, a random variable $\theta \sim \text{Beta}(\alpha, \beta)$ with $\alpha, \beta > 0$ has the density

$$p(\theta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1}(1-\theta)^{\beta-1}, \qquad 0 \le \theta \le 1.$$

The fraction in front does not depend on θ; it only makes the density integrate to 1.

Γ is the **gamma function**, which extends the factorial from the integers to all positive real numbers:

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t}\, dt, \qquad x > 0.$$

You will rarely need the integral itself. What you need is its key property, which follows from integrating by parts,

$$\Gamma(x+1) = x\,\Gamma(x), \qquad \Gamma(1) = 1,$$

so for a positive integer n it is just a factorial: $\Gamma(n) = (n-1)!$, e.g. $\Gamma(4) = 3! = 6$. For non-integer arguments, compute it numerically: `scipy.special.gamma(x)` in Python, `gamma(x)` in R. Γ grows even faster than the factorial and overflows already around $x = 172$, so for large arguments work with its logarithm instead: `scipy.special.gammaln(x)` in Python, `lgamma(x)` in R.

1. [Pen & paper] Suppose you have a $\text{Beta}(4, 4)$ prior distribution on the probability θ that a coin will yield a 'head' when spun in a specified manner. The coin is independently spun ten times, and 'heads' appear **fewer than 3 times**. You are not told how many heads were seen, only that the number is less than 3. Calculate your exact posterior density (up to a proportionality constant) for θ.

here your solution

2a. [Pen & paper] Consider the following observations: T H T H T T T T T T. Suppose you have a $\text{Beta}(\alpha, \beta)$ prior distribution on the probability θ representing the probability of getting "heads". Determine a general formula for the posterior distribution of θ.

here your solution

2b. [Pen & paper] Now generalize 2a. Keep the $\text{Beta}(\alpha, \beta)$ prior, but suppose the coin is spun n times and shows y heads. The likelihood is binomial,

$$p(y \mid \theta) = \binom{n}{y} \theta^y (1-\theta)^{n-y},$$

and as a function of θ it has the same shape as a beta density. Show that the posterior is again a beta distribution, and give its parameters.

here your solution

3. [Code] *[optional]* Plot the posterior density you derived in question 1 over $\theta \in [0, 1]$.

The prior is plotted for you below. `stats.beta.pdf(x, a=alpha, b=beta)` is the equivalent of R's `dbeta(x, a, b)`, so in R the same plot is:

```r
theta <- seq(0, 1, length.out = 1000)
plot(theta, dbeta(theta, 4, 4), type = "l",
     xlab = "Probability of Heads", ylab = "Density")
abline(v = 0.5, lwd = 3)
```

```python
ii = np.linspace(0, 1, 1000)

fig, ax = plt.subplots(figsize=(7, 4.5))
ax.plot(ii, stats.beta.pdf(ii, a=4, b=4))
ax.axvline(0.5, color="black", linewidth=3)
ax.set_xlabel("Probability of Heads")
ax.set_ylabel("Density")
ax.set_title("Probability of Heads Prior Distribution Beta(4, 4)")
fig.tight_layout()
plt.show()
```

```
# here your code
```

4. [Code] *[optional]* Using the formula derived in question 2a, plot the posterior for θ using these priors:

- $\text{Beta}(4, 4)$
- $\text{Beta}(1, 1)$
- $\text{Beta}(2, 5)$
- $\text{Beta}(8, 10)$

In R, `par(mfrow = c(2, 2))` gives the 2x2 grid that `plt.subplots(2, 2)` gives in Python, and `lines()` adds a second curve to the current panel:

```r
par(mfrow = c(2, 2))
plot(theta, dbeta(theta, 4, 4), type = "l", lty = 2, col = "green")   # one panel
lines(theta, dbeta(theta, 2, 2), col = "blue")                         # a second
curve on it
```

```
# here your code
```

Exercise 2: Mean and Variance of the Beta Distribution

A posterior is a whole distribution, but we often summarize it by its mean and variance. In exercise 3 we will need both for a beta posterior, so let us derive them here. Throughout, $\theta \sim \mathrm{Beta}(\alpha, \beta)$ with the density from exercise 1.

1. [Pen & paper] The density integrates to 1. Use this to show that

$$\int_0^1 \theta^{\alpha-1}(1-\theta)^{\beta-1}\, d\theta = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}.$$

This holds for *any* $\alpha, \beta > 0$, so it lets you do every integral of this shape without integrating.

here your solution

2. [Pen & paper] Show that the mean is

$$E(\theta) = \frac{\alpha}{\alpha+\beta}.$$

Hint: write $E(\theta) = \int_0^1 \theta\, p(\theta)\, d\theta$, notice that the integrand is again of the shape in question 1, and simplify with $\Gamma(x+1) = x\,\Gamma(x)$.

here your solution

3. [Pen & paper] Compute $E(\theta^2)$ in the same way, then use $\mathrm{Var}(\theta) = E(\theta^2) - E(\theta)^2$ to show that

$$\mathrm{Var}(\theta) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}.$$

here your solution

Exercise 3: Probability of a Girl Birth Given Placenta Previa (BDA3 p. 37)

As a specific example of a factor that may influence the sex ratio, we consider the maternal condition **placenta previa**, an unusual condition of pregnancy in which the placenta is implanted low in the uterus, obstructing the fetus from a normal vaginal delivery. An early study concerning the sex of placenta previa births in Germany found that of a total of 980 births, 437 were female. How much evidence does this provide for the claim that the proportion of female births in the population of placenta previa births is less than 0.485, the proportion of female births in the general population?

1a. [Pen & paper] Assume a uniform prior for the probability of a girl birth. What is the posterior distribution of θ? What are the mean and the variance of this posterior distribution? Use the formulas from exercise 2.

here your solution

1b. [Code] Check your mean and variance numerically.

```
# here your code
```

1c. [Code] Plot this posterior distribution.

```
# here your code
```

2. [Code + Pen & paper] Compute the 95% posterior interval of the posterior distribution, and the posterior probability that $\theta < 0.485$. What do they say about the claim that the proportion of female births is lower among placenta previa births?

> **Tip:** `stats.beta.ppf(q, a=alpha, b=beta)` is the equivalent of R's `qbeta` - it maps a quantile to a value of θ. `stats.beta.cdf(x, a=alpha, b=beta)` is R's `pbeta` - it goes the other way.

In R, with `a` and `b` the parameters of your posterior:

```
qbeta(c(0.025, 0.975), a, b)   # the two ends of a 95% interval
pbeta(0.5, a, b)               # P(theta < 0.5)
```

```
# here your code
```

here your solution

3. [Code] Add to your plot the posterior interval (shaded) and a dotted line at 48.5% for the population average.

In R, `polygon()` shades the area under a curve (matplotlib's `fill_between`) and `abline()` draws the vertical line (`axvline`). With `lower` and `upper` the ends of your interval:

```
xs <- seq(lower, upper, length.out = 100)
polygon(c(lower, xs, upper), c(0, dbeta(xs, a, b), 0), col = "lightblue", border
= NA)
abline(v = 0.485, lty = 3)
```

```
# here your code
```

4. [Code + Pen & paper] Illustrate the effect of the prior by comparing posterior distributions with different parameter values for the beta prior distribution. Which priors visibly move the posterior, and why? Use the following three choices for priors:

- $\text{Beta}(0.485 \times 2, (1 - 0.485) \times 2)$
- $\text{Beta}(0.485 \times 20, (1 - 0.485) \times 20)$
- $\text{Beta}(0.485 \times 200, (1 - 0.485) \times 200)$

All three are centered on 0.485; they differ only in how *concentrated* they are, i.e. how many pseudo-observations of prior information they represent.

```
# here your code
```

here your solution

Exercise 4: Mean and Variance of Conditional Distributions (BDA3 eqs. 1.8, 1.9)

[optional]

Let u and v be two random variables with joint density $p(u, v) = p(u \mid v)\, p(v)$. $E_u(u \mid v) = \int u\, p(u \mid v)\, du$ is the mean of u for a fixed v; it is a function of v, so we can average it over $p(v)$ in turn. The subscript names the variable that is averaged over: E_u and Var_u integrate over u, E_v and Var_v over v.

1. [Pen & paper] Show that

$$E_u(u) = E_v(E_u(u \mid v))$$

by expanding out the right-hand side.

here your solution

2. [Pen & paper] Show that

$$\text{Var}_u(u) = E_v(\text{Var}_u(u \mid v)) + \text{Var}_v(E_u(u \mid v)).$$

Hint: write both terms on the right with $\text{Var}(x) = E(x^2) - E(x)^2$, and note that question 1 holds for any function of u, in particular $E_u(u^2) = E_v(E_u(u^2 \mid v))$.

here your solution